## Emergency logistics for **wildfire** suppression based on forecasted disaster evolution

Z Yang, L Guo, Z Yang - Annals of Operations Research, 2019 - Springer

… fire **propagation model** is first built using both the flame-igniting attributes of wildfires and the factors affecting **wildfire propagation** … fire **propagation model** in this work. The Wangzhengfei …

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## An agent-based **model** to represent space-time **propagation** of **forest-fire** smoke

AK Smith, S Dragicevic - ISPRS Annals of the …, 2018 - isprs-annals.copernicus.org

… -based **model** (ABM) for the **propagation** of **forest-fire** smoke … patterns of spatio-temporal **propagation**. The developed ABM … **model** was developed using data from the 2017 **forest fire** …

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## Integrated **wildfire** danger models and factors: A **review**

I Zacharakis, VA Tsihrintzis - Science of the total environment, 2023 - Elsevier

… apply best to fire **behavior prediction**, while other techniques referenced in the current **review** are potentially useful but further investigation is needed. In conclusion, **wildfire** danger is a …

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## A **review** of optimization and decision models of prescribed burning for **wildfire** management

J Qi, J Zhuang - Risk Analysis, 2024 - Wiley Online Library

… The selected papers had to meet at least one of the three following criteria: (a) contribute to **wildfire** mitigation, (b) **model** fire **behavior** or influencing factors and their effects on fire behav…

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## Fire behaviour and smoke modelling: **model** improvement and measurement needs for next-generation smoke research and forecasting systems

Y Liu, A Kochanski, KR Baker, W Mell… - … of **wildland fire**, 2019 - CSIRO Publishing

… and safety concerns associated with **wildland fire** emissions. This **review** paper presents simulations and … The fire’s rate of **spread** is obtained from the Rothermel **model**. In its simplest …

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## Rapid **prediction** of **wildfire spread** using ensemble Kalman filter and polyline simplification

S Yoo, J Song - Environmental Modelling & Software, 2023 - Elsevier

… This paper proposes a new method for rapid **prediction** of **wildfire spread**, which employs computational **wildfire** simulations by FARSITE and assimilates the **simulation** results with …

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## **Wildfire** path **prediction spread** using machine learning

M Makhaba, S Winberg - 2022 International Conference on …, 2022 - ieeexplore.ieee.org

… The methodology of this study started with a literature **review** on the existing ML models in the **wildfire spread** domain, followed by a description of the high-level design of the two …

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## Recurrent convolutional deep neural networks for modeling time-resolved **wildfire spread behavior**

J Burge, MR Bonanni, RL Hu, M Ihme - Fire Technology, 2023 - Springer

… To assess the accuracy of the DNN models in predicting the **dynamics** of **wildfire propagation**, we measure the accuracy of each DNN **model** over a time period of 1500 min (100 …

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Some requirements for simulating **wildland fire behavior** using insight from coupled weather—**wildland fire** models

J Coen - Fire, 2018 - mdpi.com

… **Wildland fire behavior** models have been developed and … environmental conditions on fire **behavior**. Early models were … fire **behavior simulation** results to advise potential users on …

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Mathematical modeling of **wildland fire** initiation and **spread**

V Agranat, V Perminov - Environmental Modelling & Software, 2020 - Elsevier

The aim of this paper is to create a user-friendly computational tool for analysis of **wildland fire behavior** and its effect on urban and other structures. A physics-based multiphase …

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